

Ibex GP Progress

(Zero-RISC)

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Outlines

- Tech File Explained
- Routing Initial Attempts
- Routing Final Attempts
- Inquiries

Tech File Explained

Cell Dimensions

Table 2.14. Physical structure specification

Parameter	Symbol	Value
Cell height	H	0.6 μm
Power rail width	W_1	0.094 μm
Vertical grid	W_2	0.06 μm (d_{track})
Horizontal grid	W_3	0.074 μm (d_{track})
Horizontal grid offset	-	0.037 μm
NWell height	W_4	0.253 μm

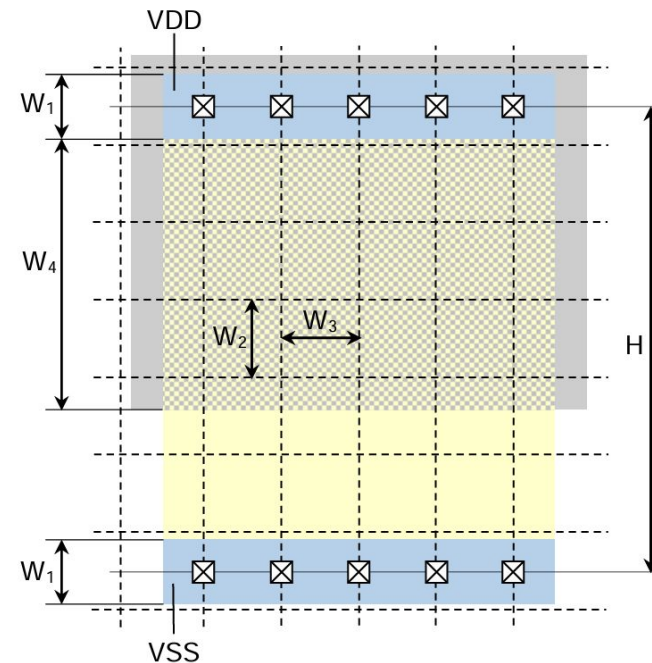


Figure 2.1. Physical structure of digital standard cells

TF Quick Comparison (M1)

Drive

PDK

Layer "M1" {		536 Layer "M1" {	
layerNumber = 19		537 layerNumber = 19	
isDefaultLayer = 1		538 isDefaultLayer = 1	
maskName = "metal1"		539 maskName = "metal1"	
numMasks = 2		540 numMasks = 2	
pitch = 0.074		541 pitch = 0.15	
visible = 1		542 visible = 1	
selectable = 1		543 selectable = 1	
blink = 0		544 blink = 0	
color = "cyan"		545 color = "cyan"	
lineStyle = "solid"		546 lineStyle = "solid"	
pattern = "dot"		547 pattern = "dot"	
minWidth = 0.034		548 minWidth = 0.034	
maxWidth = 3		549 maxWidth = 3	
defaultWidth = 0.034		550 defaultWidth = 0.034	
minSpacing = 0.026		551 minSpacing = 0.04	
minArea = 0.005		552 minArea = 0.01	
minEnclosedArea = 0.04		553 minEnclosedArea = 0.1	

TF Quick Comparison (M2)

Drive

PDK

Layer "M2" {	627 Layer "M2" {
layerNumber = 21	628 layerNumber = 21
maskName = "metal2"	629 maskName = "metal2"
numMasks = 2	630 numMasks = 2
isDefaultLayer = 1	631 isDefaultLayer = 1
visible = 1	632 visible = 1
selectable = 1	633 selectable = 1
blink = 0	634 blink = 0
color = "green"	635 color = "green"
lineStyle = "solid"	636 lineStyle = "solid"
pattern = "dot"	637 pattern = "dot"
pitch = 0.06	638 pitch = 0.3
defaultWidth = 0.034	639 defaultWidth = 0.034
minWidth = 0.034	640 minWidth = 0.034
minSpacing = 0.026	641 minSpacing = 0.04
maxWidth = 5	642 maxWidth = 5
sameNetMinSpacing = 0.026	643 sameNetMinSpacing = 0.034
minArea = 0.005	644 minArea = 0.016
minEnclosedArea = 0.04	645 minEnclosedArea = 0.1
	646

PDK Recommended Routing Scheme

2.3. NXT Architecture

In designs using NXT architecture, the preferred routing direction for M2 is parallel to the cell power/ground (P/G) rails. This routing direction is conventionally referred to as “horizontal”. The preferred routing direction for M1 is horizontal if the particular cell architecture has a M1 power/ground rail, and vertical otherwise. The preferred routing directions in the cell layouts naturally extend to routing at the block and chip level as well.

Preferred routing directions for SAED standard cell library are horizontal for M1 and M2, vertical for M3, and alternating horizontal and vertical thereafter. This is conventionally referred to as the **HHVH** routing scheme.

One of the advantages of the NXT routing architecture is that there are many M2 tracks that can access the M1 pins on a given cell. This enhances the ability of a routing tool to connect to these pins, since M2 is the primary layer used by routers to approach M1 cell pins. Improving pin access in this way minimizes congestion in the vicinity of a cell, since the router is no longer restricted to a single M2 track when connecting to a given pin.

Routing Initial Attempts

Legality Errors

Area was small and cells with routing couldn't fit together.

ErrorSet	Total	Visible
ibex_hvt:ibex_hvt_routing.design	4095	4095
ibex_hvt_routing_check_legality.err	4095	4095
Cell_Cell_Overlap	3049	3049
Offrow	537	537
Spacing	506	506
Voltage Area	3	3

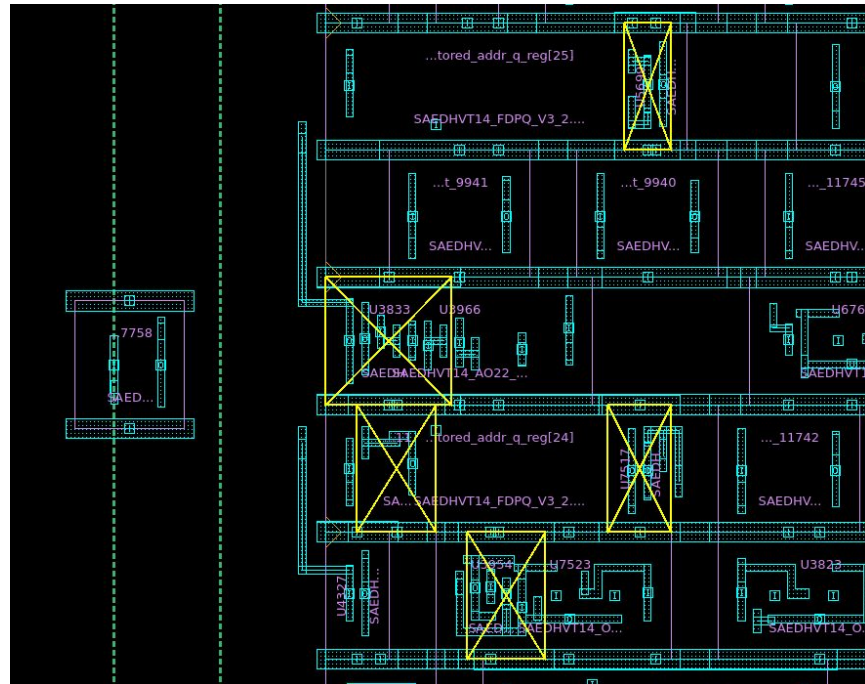
```

1 Loading cell instances...
2 Number of Standard Cells: 21936
3 Number of Macro Cells: 0
4 Number of IO Pad Cells: 0
5 Number of Blocks: 0
6 Loading P/G wires and vias...
7 Number of GND Wires: 1466
8 Number of GND Vias: 1444
9 Number of GND Terminals: 0
10 Number of VDD Wires: 1479
11 Number of VDD Vias: 1456
12 Number of VDD Terminals: 0
13 *****Verify net GND connectivity*****
14 Number of floating wires: 0
15 Number of floating vias: 0
16 Number of floating std cells: 520
17 Number of floating hard macros: 0
18 Number of floating I/O pads: 0
19 Number of floating terminals: 0
20 Number of floating hierarchical blocks: 0
21 *****
22 *****Verify net VDD connectivity*****
23 Number of floating wires: 0
24 Number of floating vias: 0
25 Number of floating std cells: 520
26 Number of floating hard macros: 0
27 Number of floating I/O pads: 0
28 Number of floating terminals: 0
29 Number of floating hierarchical blocks: 0
30 *****
31 Overall runtime: 0 seconds.

```

Legality Errors

Overlapped cells



[illegible]

Zroute Errors

After increasing area, legality errors vanished away but Zroute was still exist.

ErrorSet	Total
ibex_hvt:ibex_hvt_routing.design	1431
__ibex_hvt_cts_check_legality.err	0
__ibex_hvt_routing_check_legality.err	0
zroute.err	1431
Diff net spacing	575
Diff net var rule spacing	3
Diff net via-cut spacing	23
Edge-line via spacing	2
Less than NDR width	4
Less than minimum area	229
Off-grid	12
Same net spacing	18
Same net via-cut spacing	61
Short	504

Routing Final Attempts

Addressing Zroute Errors

Changes in Floor planning and Power planning.

```
Floorplan
```

```
-----
```

```
>> 140 sq. + 18 extention core to die - Aligned tracks - M1 vertical - clk & rst on M9
```

```
Powerplan
```

```
-----
```

```
>> Ring region offset 2 instead of 1
```

We think they might not be so effective.

Addressing Zroute Errors

Changes in Placement.

```
Placement
-----
>> set_placement_spacing_rule -labels {X X} {0 3}
>> set_app_option -name place.coarse.congestion_driven_max_util -value 0.5
>> set_app_options -name place.coarse.max_density -value 0.2
>> set_app_options -name place.coarse.target_routing_density -value 0.7
>> route_global // after place_opt
```

Increased cell padding and changing options values contributed to a better result but using route_global after place_opt had a huge impact on better routing with less DRCs.

Addressing Zroute Errors

Changes in CTS.

```
CTS
-----
>> set_input_transition -rise [expr 0.18*$CLK1_PER] [get_ports clk_i]
>> set_input_transition -fall [expr 0.1*$CLK1_PER] [get_ports clk_i]
>> create_routing_rule $CLK_NDR_NAME -multiplier_spacing 1 -multiplier_width 1
>> set_app_options -as_user_default -list {cts.optimize.enable_congestion_aware_ndr_promotion true}
```

We thought of relaxing the input transition for the clock port so we save utilization resources obtained by CTS but turns out to be not so effective in routing DRCs.

Inquiries

Q 1

Is it a common approach to use route_global after placement and before CTS and how effective it is ?

Q 2

Is it safe for the ring to be close to the core area where the cells are placed?

Q 3

Formality passed fine before in previous synthesis runs but once we source `dont_use` constraints, formality check reports many failing points, is there any explanation as that's the only one change made ?

Q 4

Should we include antenna rules in routing? If so, where is the file?

Q 5

Should we add spare cells, in case of implementing an IP and not a chip?

Thank you.